One's-complement

- Negative number: Bitwise complement positive number
  - $0011 \equiv 3_{10}$
  - $1100 \equiv -3_{10}$
- Solves the arithmetic problem

Add		Invert, add, add carry		Invert and add	
4	0100	4	0100	– 4	1011
+ 3	+ 0011	– 3	+ 1100	+ 3	+ 0011
= 7	= 0111	= 1	1 0000	– 1	1110
		add carry:	+1		
			= 0001		

- Remaining problem: Two representations for zero
  - 0 = 0000 and also –0 = 1111

Two's-complement

- Negative number: Bitwise complement **plus one**
  - $0011 \equiv 3_{10}$
  - $110\mathbf{1} \equiv -3_{10}$
- Only one zero!
- MSB is the sign digit
  - 0 ≡ positive
  - 1 ≡ negative

Add		Invert and add		Invert and add	
4	0100	4	0100	– 4	1100
+ 3	+ 0011	– 3	+ 1101	+ 3	+ 0011
= 7	= 0111	= 1	1 0001	– 1	1111
		drop carry	= 0001		

Exercise

	6-bit Binary of Magnitude	Sign and Magnitude	One's Complement	Two's Complement
-10	001010	101010	110101	110110
-14	001110	101110	1100001	1100010
-3	000011	100011	111100	111101
-17	010001	110001	101110	101111

Arithmetic’s using 1’s and 2’s complement

One's-complement (Arithmetic)

Addition of a positive number and a negative number.

Case I: When the positive number has a greater magnitude

In this case the end-around **carry will be generated** and is **added** into the final result.

	Binary	5-bit Binary
11	1011	01011
-5	101	00101

```
1 1  1
  01011
+ 11010   (1's Complement)
-------
  00101
+     1
-------
  00110
```

One's-complement (Arithmetic)

Addition of a positive number and a negative number

Case II: When the negative number is greater.

When the negative numbers is greater **no end-around carry** will be generated. The result of addition will be negative, and the **final result is obtained by taking 1's complement** of the result.

	Binary	5-bit Binary	
-11	1011	01011	1 10100 (**1's Complement**)
5	101	00101	+ 00101
			11001
			00110 (**1's Complement**)

One's-complement (Arithmetic)

When the numbers are negative

A end-around **carry will be generated** which will be **added** in sum. 1's complement of the result will give the magnitude.

	Binary	5-bit Binary
-10	1010	01010
-5	101	00101

```
  1
    10101   (1's Complement)
+   11010   (1's Complement)
-----------
    01111
+       1
-----------
    10000
    01111   (1's Complement)
```

Two's-complement (Arithmetic)

Addition of a positive number and a negative number.

Case I: When the positive number has a greater magnitude

In this case the **end-around carry will be generated** and is **discarded**. The final result is the result of addition.

	Binary	5-bit Binary
11	1011	01011
-5	101	00101

```
 1 1  11
   01011
+  11011   (2's Complement)
 -------
   00110
```

Two's-complement (Arthimetic)

Addition of a positive number and a negative number

Case I: When the negative number is greater.

When the negative numbers is greater **no end-around carry will be generated**. The result of addition will be negative, and the **final result is obtained by taking 2's complement** of the final sum.

	Binary	5-bit Binary
-11	1011	01011
5	101	00101

```
   1 1
   10101   (2's Complement)
+  00101
  ------
   11010
   00110   (2's Complement)
```

Two's-complement (Arthimetic)

When the numbers are negative

A end-around **carry will be generated** which will be **discarded**. The final result is obtained by taking 2's complement of the sum.

	Binary	6-bit Binary
-10	1010	01011
-5	101	00101

1 111

10110 **(2's Complement)**

\+ 11011 **(2's Complement)**

10001

01111 **(2's Complement)**

In summary

- Can’t infer a representation from a number
  - 11001 is 25 (unsigned)
  - 11001 is –9 (sign magnitude)
  - 11001 is –6 (ones complement)
  - 11001 is –7 (twos complement)

- 1’s complement **-> Add Carry**

- 2’s complement **-> Drop Carry**

Thanks a lot

If you are taking a Nap, **wake up**.....Lecture Over